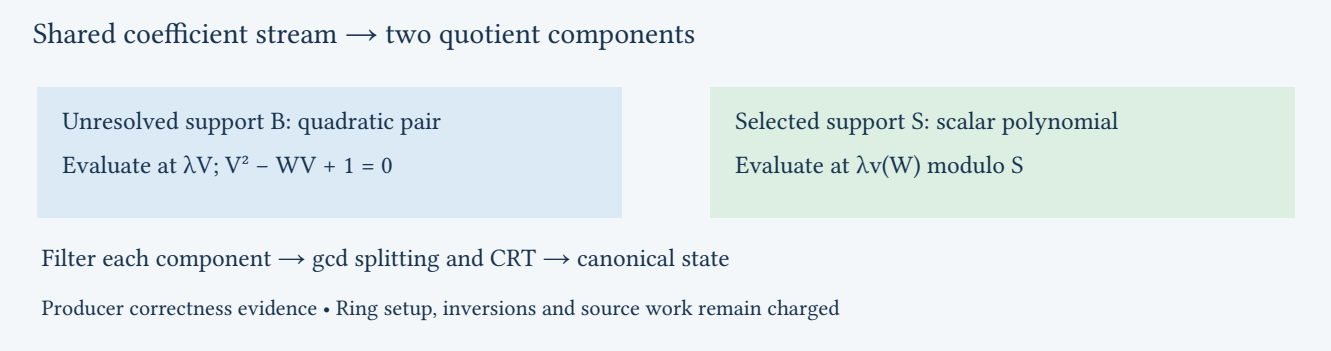
Split evaluation on canonical fibers

T-1871 · 2026-11-07 · Prime-field public-point recovery

Selected roots now use scalar quotient arithmetic; unresolved fibers retain quadratic pairs. One coefficient reader supplies both components. The state update filters each support separately and joins newly selected roots with existing roots by coprime CRT.



Correctness. Polynomial evaluation commutes with reduction modulo B and substitution V=v modulo S. Balanced composition returns the block value and base power, so each coefficient is read once even in mixed states. The canonical update retains the same candidate roots.

Native control	Observed
Mode projections	738
Mixed / selected-only / unresolved-only modes	693 / 45 / 0
Canonical states equal to reference	2,516
Public targets per cap64 / cap16 panel	281 / 281
Source calls at cap64 / cap16	2,881 / 3,152
Route changes / unresolved targets	0 / 0

Costs. For L coefficients, each present component makes 2(L−1) ring products. Scalar products invoke one base polynomial product; quadratic products invoke three. These are structural calls, not field-operation or timing ratios. Mixed states pay both components. Setup, reductions, stream production, gcds, inverse/CRT work and temporary memory remain charged. Original source grids and initial C0 remain charged.

Evidence. validation01 contains compiler/source pins and native output; all inherited checks pass. No native failure or cloud/timing trial. Independent signed verdict pending. Original 4,563-case cohort unchanged. Full details and replay command: DERIVATION.md, RESULTS.md and README.md.

Sources. Exact canonical predecessor and official FLINT polynomial quotient conventions: flintlib.org/doc/nmod_poly/index.html. KB unavailable; standard projection/CRT, no novelty claim.

Next. Remove repeated evaluator powers or share branch compilation, keeping source and initial-support costs visible.